LOCAL CONSTANCY OF THE EULER CHARACTERISTIC OF A PERFECT OBJECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A perfect object on a locally ringed space is locally represented by a finite complex of finite free modules. invoked at proof lines 8007-8010. Produces the local strictly perfect representative.
- The stalk of a finite free module has dimension equal to its rank, and the cohomology of the derived fiber is computed by the complex of vector spaces. used at proof lines 8011-8012. Identifies the value of the function on the open.

Source reference. Original lemma 0BCJ: sources/stacks-chow/tags.txt, source lines 14282–14282. Cited at proof lines 8007-8008 for the local representative; the rank computation is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $E \in D(\mathcal{O}_X)$ be perfect. The function*

$$\chi_E : X \longrightarrow \mathbf{Z}, \quad x \longmapsto \sum (-1)^i \dim_{\kappa(x)} H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \kappa(x))$$

is locally constant on X.

Proof. By Cohomology, Lemma 0BCJ we see that we can, locally on X, represent E by a finite complex $\mathcal{E}^\bullet$ of finite free $\mathcal{O}_X$-modules. On such an open the function χ_E is constant with value $\sum (-1)^i \text{rank}(\mathcal{E}^i)$. □

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